

Chapter One

The Nature of Knowledge Graphs

Introduction

A knowledge graph is a structured representation of information where entities are connected by well-defined relationships. Unlike a relational database, a graph places relationships at the center of the model.

This chapter introduces the core ideas that the rest of the book builds upon. We will move from simple triples to richer metagraph structures.

Entities and Relationships

An **entity** is any distinct thing we wish to describe: a person, a place, a concept, or an event. Each entity is given a stable identifier so that facts about it can be gathered from many sources.

A relationship connects two entities and carries its own meaning. The phrase "Alice knows Bob" is a single relationship with a clear direction and type.

TIP | Model relationships explicitly rather than hiding them inside attributes.

Reification

Reification lets us make statements about statements. When we need to attach a confidence score or a provenance record to a fact, we treat the fact itself as a node that other edges can point to.

This is the foundation for metagraphs, where an edge may connect not only entities but also other edges and entire subgraphs.

Summary

Knowledge graphs put relationships first, entities are stable and reusable, and reification unlocks statements about statements. The next chapter turns these ideas into a concrete data model.